

RAHUL KAREWAD

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Address: Bengaluru

PROFESSIONAL SUMMARY

Data Science Professional with 4.10 years of experience in data preparation, developing, and deploying highly scalable **Machine Learning models**. Hands-on experience in building, optimizing, and running ML models in production environments. Skilled in data visualization, problem-solving, and enabling data-driven decision-making.

Experienced in cloud-based solutions using AWS services including **Amazon S3, AWS Lambda**, and Amazon SES for building scalable data pipelines and automated workflows. Proficient in working **with Large Language Models (LLMs), implementing LangChain-based applications, and Agentic AI**, developing **Retrieval-Augmented Generation (RAG) systems**. Strong understanding of Transformer-based architectures and modern NLP techniques. Adept collaborator dedicated to delivering end-to-end AI solutions.

TECHNICAL SUMMARY

Programming & ML Libraries:

Python, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn

Machine Learning:

Supervised & Unsupervised Learning, Classification & Regression Algorithms, MLPipelines, ETL Pipelines, Pytorch

Algorithms:

Linear Regression, Logistic Regression, Naive Bayes, KNN, Support Vector Machines, Decision Tree, Random Forest, K-Means

Boosting Techniques:

AdaBoost, XGBoost

Deep Learning & NLP:

Neural Networks, ANN, CNN, RNN, LSTM, Activation & Loss Functions

NLP using NLTK, Gensim, SpaCy, GEN AI, **Agentic AI**

LLMs, RAG, Transformers, LangChain.

Database:

MySQL

Cloud & DevOps:

AWS (S3, Lambda, SES),

Web Framework:

Flask, FastAPI

Other Tools:

JIRA, Agile Methodology

EDUCATION

B.E – Gondwana University – CGPA: 7.59

Passing Year - 2018

HSC – Maharashtra State Board – 61.69%

SSC – Maharashtra State Board – 81.82%

PROFESSIONAL EXPERIENCE

- Grid dynamics (22 April 2026 to 21 August 2026)
 - Eli Lilly and Company (September 2025 – February 2026)
 - Fact Entry Data Solutions (May 2024 – May 2025)
 - Mphasis (April 2021 – May 2024)
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PROJECTS**Project 1: Enhancing Customer Experience with NLP-based Product Segmentation**

Domain: E-Commerce (MPHASIS)

- Developed an NLP-based product classification system to automate product segmentation and improve category accuracy in an e-commerce platform.
- Performed advanced preprocessing including tokenization, **stopword removal, lemmatization, and n-gram extraction**. Applied NLP techniques to automate and optimize product segmentation.
- **Utilized TF-IDF for keyword-based statistical importance and Word2Vec embedding for capturing semantic similarity in product descriptions to enhance classification performance.**
- Built and compared multi-class classification models, achieving high accuracy and strong macro F1-score performance.

Project 2: Fraud Monitoring System for Insurance Providers

Domain: Insurance(MPHASIS)

- **Developed the Machine Learning models** for fraud detection in insurance claims. identification of suspicious claim patterns.-scale real-time data streams.
- Developed and trained **machine learning classification models** to predict outcomes using structured data.
- Domain-specific features from claim history, transaction metadata, and behavioral indicators to improve fraud detection accuracy. Implemented ML monitoring for data drift, concept drift, and prediction accuracy.
- **Handled severe class imbalance using SMOTE**, class weighting, and threshold optimization techniques. Collected, cleaned, and preprocessed insurance claim data.
- Improved model accuracy through **hyperparameter tuning using GridSearchCV and RandomizedSearchCV cross-validation techniques**. Performed **data preprocessing, feature engineering, and model evaluation** to improve prediction accuracy.

Project 3: Customer Data Segmentation for Business Growth

Domain: Banking(MPHASIS)

- Performed customer segmentation using **K-Means clustering** and hierarchical clustering based on transaction behavior, spending patterns, and account activity.
- **Engineered RFM (Recency, Frequency, Monetary) features and behavioral metrics to identify high-value and at-risk customer segments.**
- **Applied PCA for dimensionality reduction to improve clustering stability and visualization.**
- Determined the optimal number of clusters using **the Elbow Method by analyzing Within-Cluster Sum of Squares (WCSS) to identify the point of diminishing returns. Evaluated clustering quality using Silhouette Score.**

Project 4: Automated ETL and Data Mining for Business Insights

Domain: Finance(Factentry)

- Designed and implemented scalable ETL pipelines to process large-scale structured financial datasets from multiple data sources.
- Developed Python-based data extraction scripts to process municipal corporate financial documents in identify and structure important information from textual business data for improved analytics. Applied data mining techniques for pricing behavior, maturity profiles, and coupon analysis. Stored extracted structured data into CSV/SQL databases for
- Reduced manual data extraction effort by automating parsing of financial attributes from text-based reports.

Project 5: RAG-Driven AI Chatbot for Financial Services Website

Domain: Finance (Eli Lilly and Company)

- Designed and developed a Retrieval-Augmented Generation (RAG) chatbot.
- Integrated LLMs with vector-based semantic search.
- Embedding models **GEN AI** convert text into high-dimensional numerical vectors that capture the semantic meaning of the text, enabling similarity comparison.
- Implemented document chunking, Embeddings, and similarity-based retrieval.
- Tuned prompts and retrieval thresholds to improve response accuracy.
- Reduced operational support costs through automation.
- Integrated vector databases for efficient semantic search across financial documents.
- Designed and implemented a Retrieval-Augmented Generation (RAG) architecture combining Large Language Models (LLMs) with vector-based semantic search for accurate, context-aware responses.
- Implemented semantic document chunking, embedding generation, and similarity-based retrieval to support real-time, context-aware question answering.

Project 6: Home360 CR Management & AI Document Processing System

Domain: Healthcare (Eli Lilly and Company)

- Designed and developed a **scalable FastAPI-based backend integrated with AWS S3, Lambda, and SES to automate Change Request (CR) processing workflows.**
- Implemented event-driven architecture using **AWS lambda** services to trigger automated document validation, processing, and email notifications. Designed LLM-powered summarization modules to extract key medical insights from large and complex clinical documents.
- Used **Agentic AI** to automate document processing and decision-making workflows.
- Enabled secure document storage with encryption and pre-signed URLs.
- Applied **RAG architecture** for document retrieval and summarization.
- **Performed image quality classification (High/Medium/Low).**
- Implemented CI/CD pipeline for automated testing and deployment.
- Developed multi-modal document processing for text, tables, and images using LLMs.
- Integrated vector embedding and similarity search to enable efficient semantic retrieval across healthcare documents.
- **Embeddings** → **GEN AI** Models that **convert text into numerical vectors** so machines can understand similarity between texts.

PERSONAL DETAILS

Name: Rahul Karewad

Date of Birth: 16-03-1996

Nationality: Indian

Languages Known: English, Marathi, Hindi

Current Address: Bengaluru

Date: 2026

Place: Bengaluru